

APPLICATION OF FULL STATE FEEDBACK CONTROL ON A NON-LINEAR BRAKING SYSTEM

Abstract – This paper outlines the application of full-state feedback control on an electric vehicle (EV) braking in a non-regenerative mode. It demonstrates the limitations of linear control methods in effectively controlling a nonlinear system. To address this, input-output feedback linearisation is used, transforming a nonlinear system into a linear one, enabling linear control methods to be effective. A Simulink model is developed to simulate the electromechanical and dynamics subsystems, incorporating slip ratio, braking torque, and vehicle velocity as state variables. Simulation results demonstrate that the linearised system satisfies predefined performance criteria highlighted in the report. The findings highlight that feedback linearisation offers a variable control strategy for an efficient braking system for EV, in situations where regenerative braking is unavailable.

1. INTRODUCTION

Linear control provides a wide variety of powerful tools for system analysis and control. The drawback of linear control is that it acts on the assumption of a small range of operation, where system dynamics remain approximately linear, for linear control methods to remain valid [1], [2]. In cases where the range of operation is large the performance of linear controllers drops drastically, becoming ineffective and unstable, on account of the non-linearities in the system. Traditional methods exist to address nonlinear control [3], [4], [5] but they often involve high computational cost and complexity, making them less practical for real-time applications such as a EV braking systems. EV braking systems exhibit nonlinear behaviour due to factors such as varying friction coefficients, road conditions, and dynamic interactions between braking torque and vehicle speed [6]. These nonlinearities make conventional linear controllers ineffective in maintaining stability and achieving desired braking performance across different conditions.

This paper applies feedback linearisation to an EV braking system to enable the use of conventional linear control methods. The system's performance is evaluated through simulation, assessing its ability to maintain stability and meet predefined control objectives. Section 2 of this paper describes the system dynamics during braking and section 3 highlights the model performance criteria and assumptions. Section 4 focuses on the linearisation of state equations describing system dynamics and section 5 will address simulation of the transformed linear model. Section 6 is focused on analysis of results and discussion.

2. SYSTEM DYNAMICS

The braking system is composed of two sub systems, electromechanical and system dynamics. The electromechanical subsystem is responsible for generating the braking torque, which is then applied to

the wheel to decelerate the vehicle. The equivalent circuit of the DC motor used in this subsystem is shown below:

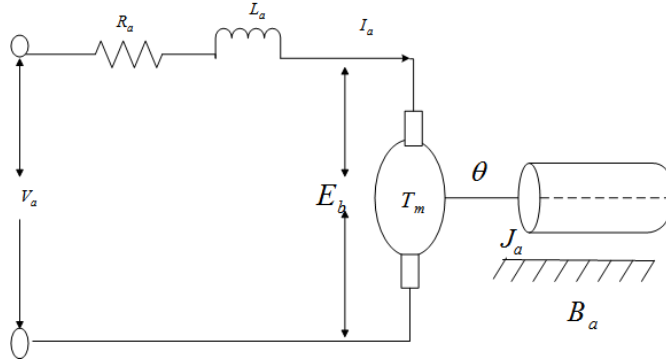


Figure 1: Equivalent circuit model of DC motor [7].

The equations that describe the equivalent circuit of the DC motor shown in figure 1 are provided below:

$$V_a = I_a R_a + L \frac{di_a}{dt} + E_b \quad (1)$$

$$E_b = K_e \omega \quad (2)$$

$$T_m = K_t i_a = \tau_b = \quad (3)$$

Developing a highly detailed wheel model introduces unnecessary complexity for a control method that aims to remain simple and efficient. Therefore, a simplified approach is adopted, focusing only on critical observable dynamics. This paper uses a quarter car model, as employed in [7], [8], [9], which reduces the systems degree of freedom to the longitudinal velocity of the vehicle and rotational speed of the wheel. This model is sufficient for analysing wheel dynamics and is illustrated in the following figure.

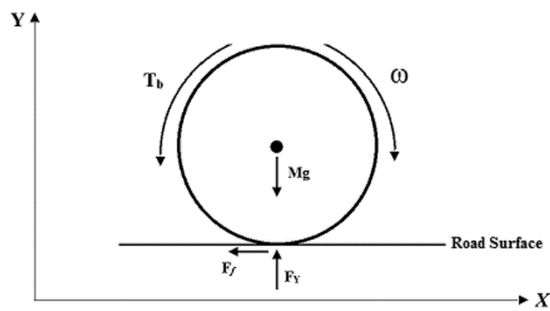


Figure 2: Free body diagram of wheel braking [10].

The first degree of freedom allows us to analysis the free body diagram in figure 2 with respect to the vehicle's longitudinal velocity. Using Newtonian laws, the corresponding equation of motions for longitudinal velocity is derived as follows:

$$M\dot{v} = -\mu N - C v^2 \quad (4)$$

where M is the mass of the quarter car model, μ is the friction coefficient, N is the normal force acting on the vehicle wheel, C is the vehicle aerodynamic friction coefficient and v is the longitudinal velocity of the car. The second degree of freedom for the quarter car model is the wheel's rotational speed. The corresponding equation for rotational speed is derived as:

$$J\dot{\omega} = \mu Nr - B\omega - \tau_b \quad (5)$$

where J is the moment of inertia, ω is the wheel rotational speed, r is the radius of the wheel, B is the wheel bearing friction coefficient and τ_b is the braking torque. The braking performance of a vehicle is governed by complex dynamic interactions between the wheel and the road surface as shown in figure 1. Central to this behaviour is the slip ratio a non-linear variable that is shown by the equation:

$$s = \frac{v - r\omega}{v} \quad (6)$$

quantifying the relative difference between the wheel speed and vehicle speed. This interaction influences the friction coefficient at the tyre road interface, which ultimately determines the effective braking force. At a slip ratio of 0, the wheel is rolling without slipping meaning the wheel speed matches the vehicle speed. In this state, there is no relative motion at the tyre road interface, so no resistive force is generated and the braking torque is effectively zero. At the other extreme, a slip ratio of 1 indicates that the wheel has completely stopped rotating while the vehicle continues moving forward. The tyre is skidding as the entire contact patch is sliding against the road surface, the friction coefficient under full slip is typically lower than the peak value achieved at optimal slip ratio, resulting in reduced braking efficiency.

The equation that defines the friction coefficient is given by:

$$\mu = 2\mu_o \frac{s_o s}{s_o^2 + s^2} \quad (7)$$

representing the ratio of tangential and the normal forces.

3. PERFORMANCE CRITERIA AND ASSUMPTIONS

The efficiency of the control method was evaluated based on parameters derived from [11], on the braking characteristics of a electric vehicle. The EV used by Kolla *et al* [11] is a Telsa Model S and provided its 5 star Euro NCAP safety rating [12], it is recognised as having an excellent braking system. The table below presents the performance criteria to be targeted.

Table 1: Performance Criteria for the simulation.

Parameter	Value
Initial Speed (km / hr)	100.91
Breaking Distance (m)	41.45

Breaking Time (s)	3.24
Braking deceleration (ms ⁻²)	9.48

These parameters are derived from an intensive braking (emergency braking) scenario on a dry road surface, allowing for the determination of the peak values of s_o and μ_o values for equation (8). The figure below allows us to determine the peak slip and friction values.

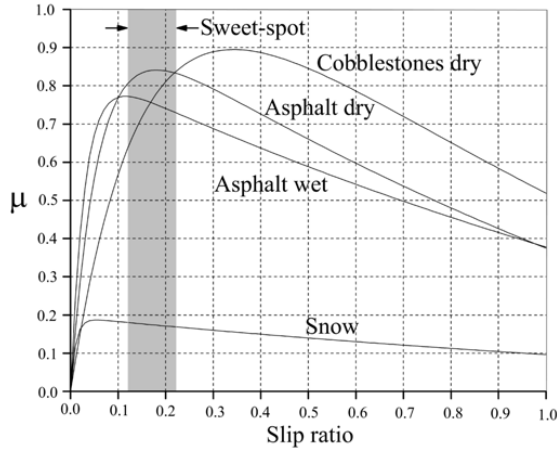


Figure 3: Friction coefficient vs slip ratio curves for multiple road surfaces [13].

The peak values define the best braking performance for any road surface and are derived from the highlighted sweet spot region in figure 3. The peak the slip value that will be used by this paper is s_o is 0.20 and the friction coefficient constant is μ_o is approximately 0.83. Due to the direct relationship between the value of slip and the braking torque performance the applied control method aims to apply a variable braking torque that maintains a slip ratio of 0.20 during braking. This ensures that the car will achieve maximum deceleration, bringing it to a stop in the shortest possible time while maintaining stability and control.

4. LINEARIZATION OF STATE EQUATIONS

The state equations that describe a non-linear system are governed by the equations:

$$\dot{x} = f(x) + gu(t) \quad (8)$$

$$y = h(x) \quad (9)$$

where $\dot{x}$ is the derivative of the state vector, $f(x)$ represents the natural dynamics of the system, $g(x)$ a function describing how the input $u(t)$ affects the state evolution and $h(x)$ the output of the system that we can measure. The state vector, x represents measurable properties within a system that when known, fully describe the current state of a system, this allows us to fully predict the future behaviour of a system. The quarter car model limits the analysis to translational velocity and rotational speed, which

are selected as the first two state variables. Since the torque produced by the electromechanical sub system directly affects braking performance, the input voltage to the DC motor is used as the input $u(t)$. The state vector defining the braking model is derived from equations (1), (4) and (5):

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{M}(-\mu N - C v^2) \\ \frac{1}{J}(\mu N r - B \omega - \tau_b) \\ \frac{1}{L}(V_{in} - iR - K_b \omega) \end{bmatrix} \quad (10)$$

5. FEEDBACK LINEARISATION IMPLEMENTATION

Feedback linearisation aims to produce an input equation $u(t)$ that cancels the nonlinearities within the system dynamics. This is achieved by repeatedly differentiating the output of the system output $y = h(x)$ until the control input $u(t)$ explicitly appears. Through this process, a transformed system with a new state variable x_{new} and a new input until the input u_{new} is formed. The input u_{new} can then be designed using conventional linear control techniques such as pole placement to ensure system stability and achieve desired performance

The selected output $y = h(x)$ chosen for this paper is slip. The first derivative of slip with respect to time is:

$$y = \frac{v - r\omega}{v} = x_{1_{new}} \quad (11)$$

$$\dot{y} = \frac{r\omega\dot{v} - r\omega v}{v^2} = x_{2_{new}} \quad (12)$$

The input does not show up in equation 12 as it appears in the expression $\dot{\omega}$ therefore the next derivative is to be calculated:

$$\ddot{y} = \frac{r}{v^3}(-v^2\dot{\omega} + v\omega\dot{v} + 2v\dot{v}\dot{\omega} - 2\omega\dot{v}^2) = u_{new} \quad (13)$$

The input voltage term is then found in equation 13, this can be shown in the equations below:

$$\ddot{\omega} = \dot{\mu}Nr - B\dot{\omega} - K_t \frac{di}{dt} \quad (14)$$

where the factor multiplying K_t is shown in equation 10 as the third state derivative and it contains the input voltage as shown below:

$$\ddot{\omega} = \dot{\mu}Nr - B\dot{\omega} - \frac{K_t}{L}(V_{in} - iR - K_b \omega) \quad (15)$$

$$u = V_{in} = -\frac{L}{K_t}\left(\ddot{\omega} - \frac{Nr}{J}\dot{\mu} + \frac{B}{J}\dot{\omega}\right) + iR + K_b \omega \quad (16)$$

Equation 16 represents the equation that will cancel the nonlinearities in the system transforming it to a linear system that can be controlled. The addition of equation 13 within equation 16, then allows for

pole placement to be implemented in our new linearised model. This can be done by making $\ddot{\omega}$ the subject of the formular in the equation 13, this simplifies to below to become:

$$\ddot{\omega} = \frac{1}{v^2} (-(\frac{v^4}{r} u_{new} - v\omega\ddot{v} - 2vv\dot{\omega} + \dot{2}\omega\dot{v}^2)) \quad (17)$$

The expression for u_{new} that allows us to perform pole placement is expressed as:

$$u_{new} = Kx_{new} \quad (18)$$

Where x_{new} represents the new state variables x_{1new} and x_{2new} derived in equation 11 and 12.

5. SIMULATION RESULTS AND EVALUATION

The Simulink model that was used for the simulation of state dynamics is shown in appendix A. The following graphs show the difference of the slip output in the non-linearised model and the model where linearisation is implemented.

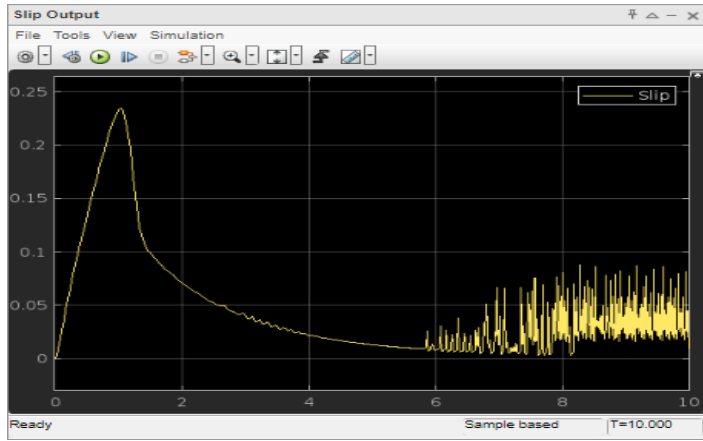


Figure 4: Slip Output before linearisation.

As shown by figure 4, the system shows a nonlinear output to a unit step input in simulation. The implementation of equation 16, without the variable of u_{new} allows us to see the effect of linearisation on the nonlinear model.

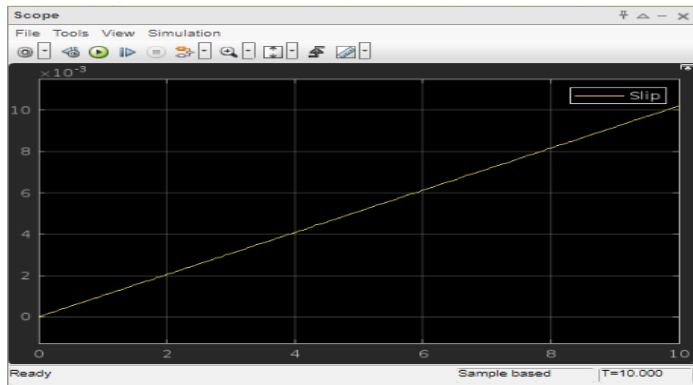


Figure 5: Slip output from linearised model without control.

Figure 5 shows that the slip now produces a linear relationship with a sample input. This makes us know that the linearisation effect of the control input works. The application of a control method using pole placement allows us to ensure that the system works in line with the performance criteria in table 1. The control law that is used is $u_{new} = r - Kx_{new}$, where r is used as a reference slip of 0.2 to ensure that if effective pole placement is done the slip saturates at 0.20. A gain Kx_{new} matrix is created multiplying two gains K_1x_{1new} and K_2x_{2new} , where x_{new} are the new state variables from the differentiation process, these can be found in equation 11 and 12. After the implementation of the reference control law the resultant slip produced by the model is presented below:

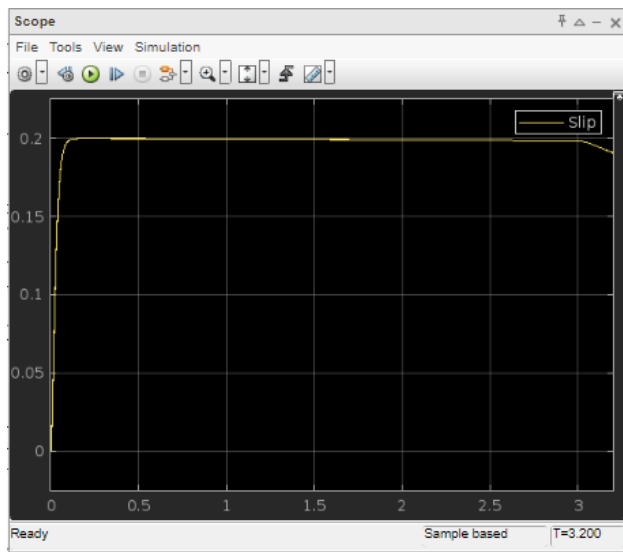


Figure 6: Slip versus time output using linear control law.

Figure 6 shows the output when a control law is applied to the model, as can be seen by the figure it saturates at a slip of 0.2 showing that optimal slip control has been achieved and at 3 seconds the simulation fails due to the vehicle having stopped.

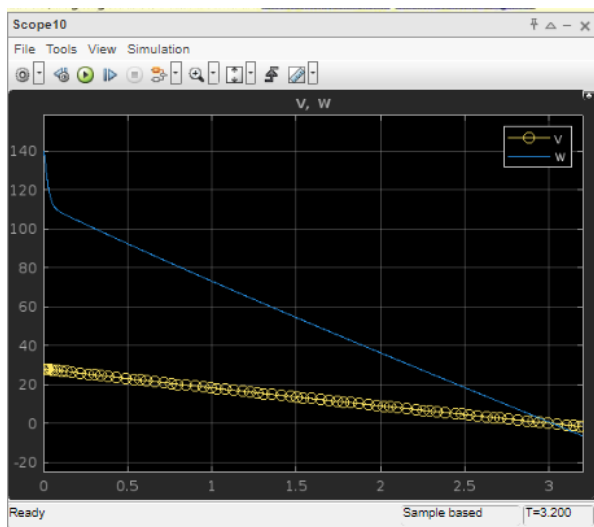


Figure 7: Vehicle velocity and wheel velocity versus time for linearised system.

Figure 7, shows that the velocity highlighted by the green line reduces achieving a value of zero at the 3 second mark from the initial 29 ms^{-1} , which is a better performance variable compared to the 3.24 seconds in table 1. The wheel velocity also reaches the zero after 3 seconds from an initial wheel velocity of 140 rads^{-1} , confirming that the car does stop at the 3 second mark. The deceleration achieved by the vehicle is 9.402 ms^{-2} , the value from the gradient of the velocity versus time plot the calculation shown in appendix B from a Simulink screen capture. Using these parameters and the equations below:

$$s = ut + \frac{1}{2}at^2 \quad (19)$$

$$s = (29)(3) + \frac{1}{2}(-9.402)(3)^2 = 41.691m \quad (20)$$

where u is the initial velocity, t is the time taken to stop and a is the deceleration rate. The calculated distance stopped is 41.681 meters which is almost the value highlighted in table 1.

CONCLUSION

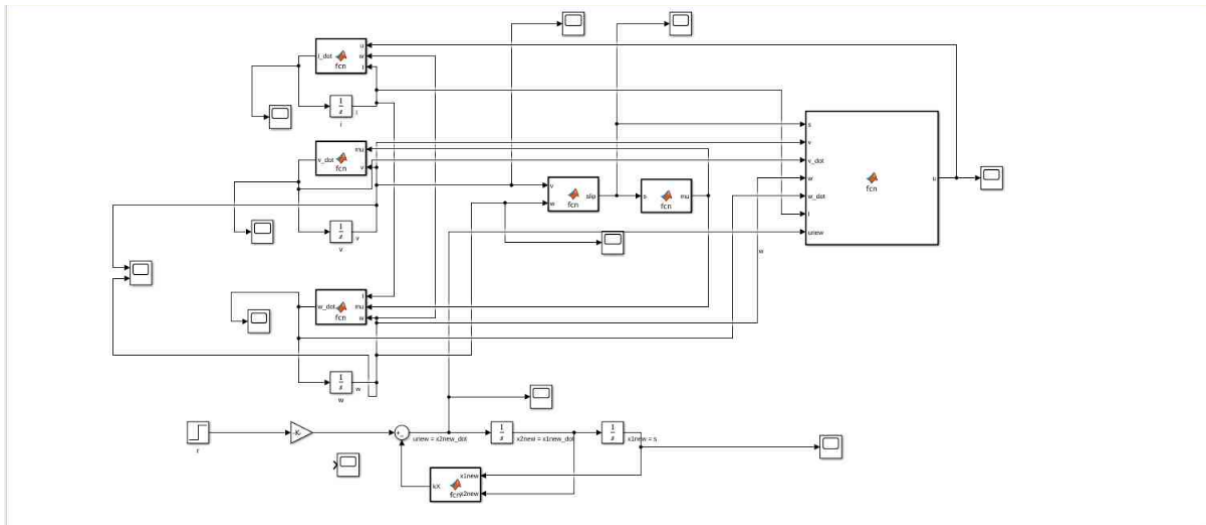
The implementation of the feedback linearisation technique successfully enabled slip control for a nonlinear braking system. Through the transformation of the nonlinear system into a linear and controllable form, it enabled for the design of an effective control law with pole placement to meet desired performance. The system met its stopping time, deceleration and distance criteria demonstrating the effectiveness of feedback linearisation in achieving accurate and stable braking control in nonlinear vehicle systems.

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Appendix

Appendix A: Outline of Simulink model for the system model



Appendix B: Calculation of rate of deceleration

